EXPLAINABLE AI · CLINICAL AI · EEG

H-CBM -EEG

Hierarchical Concept Bottleneck Model for Trustworthy EEG-to-Emotion LLMs

Domain: Explainable AI + Clinical AI + EEG Emotion Recognition

Motivation: Building clinically trustworthy AI that not only predicts emotions from EEG signals but also explains why.

Research Problem Clarity

EEG-to-LLM systems can predict emotions and explain them from brain signals, yet three critical challenges remain.

01

Hallucinated Explanations

- Explanations may sound convincing
- Not guaranteed to be grounded in EEG evidence
- No way to verify faithfulness

02

Black-Box Architecture

- Opaque MLP projection layer
- Clinicians cannot see intermediate reasoning
- No interpretable psychological concepts

03

No Human Validation

- Only automated metrics used
- No expert trustworthiness evaluation
- Limited clinical applicability

RESEARCH GAP

No existing EEG-LLM framework provides all three at once, a gap identified in current EEG-LLM literature.

- **Interpretable reasoning**
- **Faithfulness verification**
- **Human-in-the-loop validation**

Proposed Idea: H-CBM-EEG

EEG Signal → EEGPT Encoder (frozen)

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Level 1 · Neurophysiological Alpha · Theta · Gamma · Asymmetry

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Level 2 · Affective Valence · Arousal · Dominance

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Level 3 · Cognitive Appraisal Novelty · Goal · Coping

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Confidence Estimator P(concept)

↓

Qwen LLM high confidence | **Human Reviewer** low confidence

↓

Emotion Prediction + Explanation

Key Innovation

Replace the opaque projection layer with a 3-level Hierarchical Concept Bottleneck inspired by Scherer's Component Process Model.

Human-in-the-Loop

- Experts can correct concepts before the explanation
- Low-confidence cases are escalated

Why Novel? First system combining:

- EEG + Hierarchical CBM
- Faithfulness verification
- Human-in-the-loop
- Human-judge evaluation

Dataset / Simulation

Datasets Used

Dataset	Purpose
SEED-VII	Primary training dataset
DEAP	Cross-corpus testing
DREAMER	Additional validation

Human Evaluation

~150 explanation pairs · 3+ expert evaluators · blind comparison

SEED-VII Statistics

20	Subjects
7	Emotion classes
62	EEG channels
24,444	Training segments

Same dataset as E²-LLM, so the baseline comparison is fair.

SIMULATION 1

Cross-Corpus Generalization

Train on SEED-VII, test on DEAP / DREAMER

SIMULATION 2

Counterfactual Perturbation

Flip positive to negative valence; check whether the explanation changes accordingly

SIMULATION 3

Ablation Study

Flat projection vs 1-, 2-, 3-level CBM: contribution of each level

Comparison with SOTA Idea and Expected Result

Aspect	E²-LLM (SOTA)	H-CBM-EEG (Ours)
Projection	Flat 2-layer MLP + GELU, opaque	**3-level hierarchical concept bottleneck**
Interpretability	Intermediate representation has no meaning	**Concepts at 3 levels (neuro, affect, appraisal)**
Faithfulness	Chain-of-thought only, never verified	**FaithScore via counterfactual perturbation**
Human validation	None (stated limitation)	**Blind human judges, 3+, Cohen's Kappa**
Oversight	None	**Clinician intervention + confidence escalation**
Evaluation data	SEED-VII only (20 subjects)	**SEED-VII → DEAP / DREAMER**

EXPECTED RESULT

RQ1 Accuracy is comparable or better, with improved cross-corpus generalization over the flat projection.

RQ2 Higher FaithScore: explanations change when a concept is flipped.

RQ3 Higher human ratings for trust and clinical use; low confidence flags wrong predictions.

Research Impact

Our 4 Core Contributions

1 Hierarchical Concept Bottleneck

Raw EEG features become structured psychological concepts.

2 FaithScore (New Metric)

Faithful explanation changes ÷ total perturbations.

3 Dual Human-in-the-Loop

Clinician intervention and confidence escalation.

4 Human-Judge Evaluation

Experts rate faithfulness, plausibility, clarity, actionability.

Scientific Impact

- First hierarchical CBM for EEG-LLM systems
- First FaithScore benchmark
- First human-validated EEG-LLM framework

Clinical Impact

- Transparent AI decisions
- Human oversight
- Increased trust in neuro-AI systems

FUTURE VISION Towards clinically trustworthy Brain-AI systems that explain, justify, and verify every emotional prediction.